

Xiangyu Xie

Ph.D. Student at USTC

Advisor: Prof. Jun Jiang | Sep 2026 – Present

AI for Science | Molecular Machine Learning | Scientific Agents

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Education

University of Science and Technology of China (USTC)

Ph.D. Student | Sep 2026 – Present

Advisor: Prof. Jun Jiang

Undergraduate studies in Chemical Physics | From Fall 2022

Undergraduate thesis completed 1 May 2026; supervised by Prof. Jun Jiang.

Selected Research & Projects

QuantumAgent

2026

Evidence-Grounded Quantum Physics Teaching Agent

Independent Developer | Completed competition prototype

Second Prize, Graduate Division | Agent Track, USTC “107 Cup” (Sep 2026) — Solo entrant.

- Inspired by my Spring 2026 Quantum Physics teaching assistantship, independently built a teaching agent with a React/TypeScript workbench and a FastAPI/LangGraph backend, integrating course-grounded retrieval and scientific tools.
- Implemented source-traceable retrieval and a generated-code pipeline with AST checks, isolated execution, and independent numerical validation for a finite rectangular barrier case study.
- Developed staged learning interactions spanning initial attempts, diagnosis, teach-back, transfer, and independent problem solving, with persistent learning records and recoverable workflow state.

EGNN for RTP Property Prediction

Fall 2025 – Spring 2026

Undergraduate Thesis | Advisor: Prof. Jun Jiang

- Built a 3D molecular graph pipeline for approximately 6,000 organic room-temperature phosphorescence (RTP) candidate molecules with quantum-chemistry labels.
- Implemented an eight-layer EGNN with 21-dimensional atom features and six-dimensional bond features. Compared with a character-level SMILES Transformer on the same 4,800/1,050/150 train/validation/test split.
- Reported test-set R^2 on **150 molecules**: 0.8781 for ΔE_{S1-T1} ; 0.6263 for $SOC_{T1 \rightarrow S1}$; 0.9855 for $SOC_{T1 \rightarrow S0}$; and 0.7844 for $f_{osc,S1}$.
- Reduced test RMSE versus the lightweight SMILES Transformer by 56.1%, 29.8%, 32.2%, and 25.9%, respectively; singlet–triplet gap RMSE was 0.1248 eV. Targets describe isolated-molecule excited-state properties.

Selected Research & Projects (continued)

Multiscale Molecular Simulation

Fall 2024 – Spring 2025

Undergraduate Competition Project | Digital Intelligence Track

- Built an experiment-and-computation workflow combining GROMACS molecular dynamics with Gaussian 16 DFT calculations for n-butanol/cyclohexane systems (17:400 and 1:400).
- Sampled 100 trajectory frames from the dilute system for DFT calculations; obtained a mean dipole moment of 1.78 D, a 6.0% deviation from the measured 1.68 D.
- Used molecular environments to interpret infinite-dilution extrapolation in a physical chemistry teaching experiment; received First Prize, Central China Division, in 2025.

Doublet-Emissive Molecular Design

Fall 2023 – Fall 2024

Undergraduate Research | Supervisor: Prof. Linsong Cui

- Designed and synthesized TTM-AnQ, coupling a TTM radical framework with a 9,10-anthraquinone acceptor through Friedel–Crafts chemistry, Suzuki coupling, and oxidation.
- Performed UV–Vis, fluorescence, and cyclic voltammetry characterization; observed an emission shift from 580 to 695 nm. Completed the Undergraduate Research Program report on 12 Dec 2024.
- Estimated SOMO energies of -2.82 and -3.18 eV for TTM and TTM-AnQ, respectively, supporting an exploratory study of acceptor-functionalized radical emitters.

Awards and Honors

- **USTC Outstanding Teaching Assistant** – Spring 2026, ranked 1st (awarded Sep 2026).
- **Second Prize, Graduate Division, Agent Track** – USTC “107 Cup” Computing Power and Agent Development Competition; QuantumAgent, sole team member (Sep 2026).
- **First Prize, Central China Division** – 5th National Undergraduate Chemistry Experiment Innovation Design Competition, Digital Intelligence Track (2025).
- **USTC Outstanding Student Scholarship** (2024, 2025).
- **USTC Strengthening Basic Disciplines Plan Grant** (2025).

Teaching Experience

Teaching Assistant, USTC

- **Quantum Physics** – Spring 2026. Ranked 1st, USTC Outstanding Teaching Assistant; this teaching experience directly inspired QuantumAgent.
- **Mathematical Physics Equations (B)** – Spring 2025. Facilitated problem-solving sessions and developed supplementary materials.
- **Complex Functions (for Chemistry)** – Fall 2024. Led discussion sections, graded assignments, and provided one-on-one tutoring.

Technical Skills

Molecular learning: Python, PyTorch Geometric, RDKit, EGNs, Transformer baselines.

Scientific software: React/TypeScript, FastAPI, LangGraph, PostgreSQL, Git, Linux.

Chemistry & analysis: PACKMOL, GROMACS, Gaussian, DFT/TD-DFT, NumPy, SciPy, spectroscopy.

Research communication: LaTeX, scientific visualization, teaching materials.